

$$\textcircled{1} X \propto Y \cdot Z$$

$$\Rightarrow \frac{X}{Y \cdot Z} = k$$

$$\Rightarrow \text{new } x = ?, \text{ new } y = 2 \cdot y, \text{ new } z = 3 \cdot z$$

$$\Rightarrow \frac{\text{new } X}{\text{new } y \cdot \text{new } z} = k$$

$$\Rightarrow \text{new } X = k \cdot \text{new } y \cdot \text{new } z$$

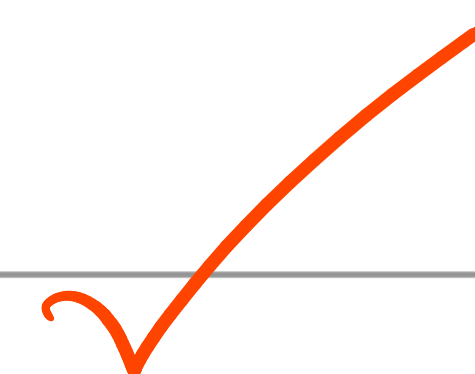
$$= \frac{X}{Y \cdot Z} \cdot 2 \cdot y \cdot 3 \cdot z$$

$$= \frac{X \cdot 2 \cdot 3 \cdot y \cdot z}{Y \cdot Z}$$

$$= \frac{6 \cdot X \cdot y \cdot z}{Y \cdot Z}$$

$$= 6 \cdot X \cdot \frac{y \cdot z}{Y \cdot Z}$$

$$= 6 \cdot X$$



n: new

$$\textcircled{2} \quad a \propto \frac{b}{c}$$

$$\frac{a \cdot c}{b} = k$$

$$na = ?, \quad nb = 2 \cdot b, \quad nc = c \cdot 2$$

$$\frac{na \cdot nc}{nb} = k$$

$$na = k \cdot nb \cdot \frac{1}{nc}$$
$$= \frac{a \cdot c}{b} \cdot 2 \cdot b \cdot \frac{1}{c \cdot 2}$$

$$= \frac{1 \cdot 2 \cdot a \cdot b \cdot c}{2 \cdot b \cdot c}$$

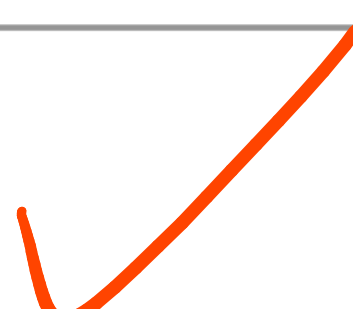
$$= 1 \cdot \frac{2}{2} \cdot \frac{b}{b} \cdot \frac{c}{c} \cdot a$$

$$= 1 \cdot a$$

Use clear variable names.

Does nb mean $n \times b$? No.

Use something
 a_1, b_1, c_1 for
the newer variables,



$$\textcircled{3} Q = \frac{k \cdot m}{p}$$

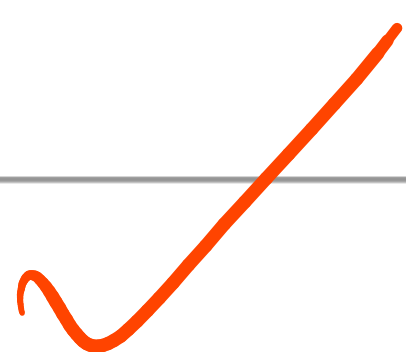
$$nQ = ? \quad nm = m, \quad np = \frac{1}{2} \times p$$

$$nQ = \frac{k \cdot nm}{np}$$

$$= \frac{k \cdot m}{(p \times \frac{1}{2})}$$

$$= \left(\frac{k \cdot m}{p} \right) \cdot \left(\frac{1}{\frac{1}{2}} \right)$$

$$= Q \cdot 2$$



$$\textcircled{4} x = 2$$
$$y \propto x^2$$

$$\Rightarrow \frac{y}{x^2} = k$$

$$\Rightarrow y = k \cdot x^2$$

$$\Rightarrow y = k \cdot 4$$

$$ny = ?, \quad nx = 6$$

$$\Rightarrow ny = k \cdot nx^2$$

$$= k \cdot 36$$

$$\frac{ny}{y} = \frac{k \cdot 36}{k \cdot 4}$$

$$= \frac{36}{4}$$

$$= 9$$



$$\textcircled{5} Z = k \cdot m \cdot h$$

$$nZ = ? \left\{ \begin{array}{l} nM = m + \frac{20}{100} \cdot m \\ \quad = m + 0.2 \cdot m \\ \quad = (m \cdot 1) + (0.2 \cdot m) \\ nm = 1.2 \cdot m \end{array} \right. \left\{ \begin{array}{l} nh = h - \frac{20}{100} \cdot h \\ \quad = h - 0.2 \cdot h \\ \quad = (h \times 1) - (0.2 \cdot h) \\ \quad = 0.8 h \end{array} \right.$$

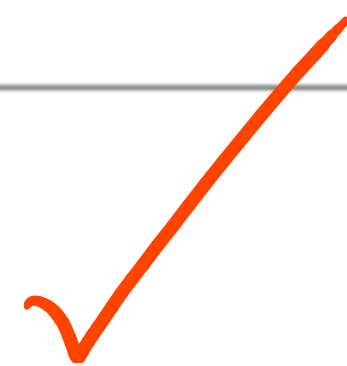
$$nZ = k \cdot nm \cdot nh = k \times (1.2)m \times (0.8)h$$

$$= \frac{12 \times 8}{100} \times k \times m \times h$$

$$= \frac{96}{100} \times k \times m \times h$$

$$= 0.96 \times Z$$

Answer: no



⑥ speed $v = 50$ mph

$$d = v \cdot t$$

distance, $d = 100$ m (miles)

$$d = v \cdot t$$

$$d = 100 \text{ m}, v = 50 \text{ mph}, t = 2 \text{ hr}$$

$$t = \frac{d}{v} = \frac{100}{50} = 2 \text{ h}$$

$$\Rightarrow nt = \frac{1}{2} \times 2 \text{ hr} \mid ns = ? \mid nd = d = 100 \text{ m}$$
$$= 1 \text{ hr}$$

$$\Rightarrow nd = nv \cdot nt$$

$$\Rightarrow nv = \frac{nd}{nt}$$

$$= \frac{100}{1}$$

$$= 100 \text{ mph}$$



⑦ time, $t = 2 \text{ hr}$

$$d = v \cdot t$$

Car A, $v_A = 60 \text{ mph}$

Car B, $v_B = 30 \text{ mph}$

$$d_A = v_A \cdot t$$

$$\begin{aligned} d_A &= 60 \times 2 \text{ m} \\ &= 120 \end{aligned}$$

$$d_B = v_B \cdot t$$

$$\begin{aligned} d_B &= 30 \times 2 \\ &= 60 \text{ m} \end{aligned}$$

Car A covers more distance.

$$d_A : d_B = 120 : 60 = 2 : 1$$

$$v \propto \frac{1}{t}$$

⑧ new $d = d$, $nv = 3 \times v$, $nt = ?$

$$d = v \cdot t$$

$$n \quad d = nv \times nt$$

$$\begin{aligned} nt &= \frac{nd}{nv} = \frac{d}{3 \times v} \\ &= \frac{v \times t}{v \times 3} \\ &= \frac{t}{3} \end{aligned}$$

The trip will take $\frac{1}{3}$ rd the previous time

$$\textcircled{9} \quad m = 10 \text{ kg}$$

$$a = 2 \text{ m/s}^2$$

$$F = m \cdot a$$

$$= 10 \times 2$$

$$F = 20 \text{ N}$$

$$nm = 20 \text{ kg}$$

$$nF = F = 20 \text{ N}$$

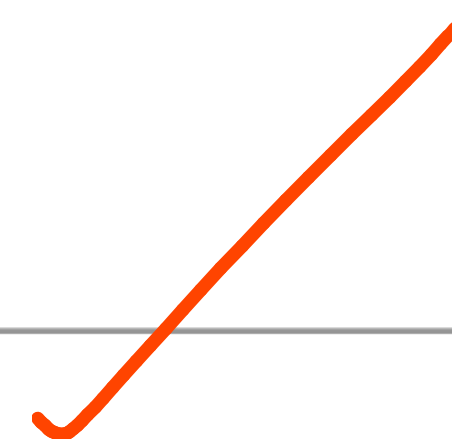
$$na = ?$$

$$nF = nm \cdot na \quad F = m \cdot a$$

$$na = \frac{nF}{nm}$$

$$= \frac{20}{20}$$

$$= 1 \text{ m/s}^2$$



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$$F = ma$$

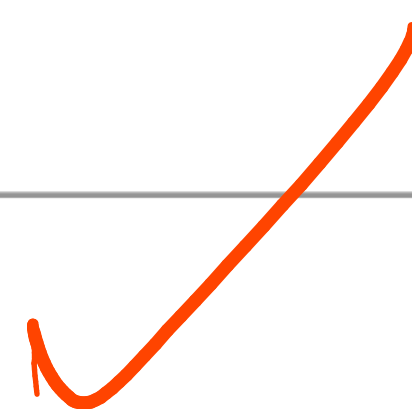
$$nF = ?$$

$$nm = 2m, na = a$$

$$nF = nm \cdot na$$

$$= 2m \cdot a$$

$$= 2F$$



①①

$$F = G \frac{m_1 \cdot m_2}{r^2}$$

$$n r = 3 r,$$

$$n r \neq n v$$

$$n m_1 = m_1$$

$$n m_2 = m_2$$

$$n F = ?$$

$$n F = G \frac{n m_1 \cdot n m_2}{n r^2}$$

$$= G \frac{m_1 \cdot m_2}{(3r)^2}$$

$$= \frac{1}{9} \times G \frac{m_1 m_2}{r^2}$$

$$= \frac{1}{9} \times F$$



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$$m = m$$

$$v = 10 \text{ m/s}$$

$$K.E. = \frac{1}{2} \cdot m \cdot v^2$$

$$= \frac{1}{2} \cdot 10^2 \cdot m$$

$$= \frac{1}{2} \times \underline{40} \cdot m = \frac{1}{2} \times 100 \times m$$

$$= \underline{20} \cdot m = 50 \text{ m}$$

$$N m = m, \quad N v = 40 \text{ m/s},$$

$$n K.E. = \frac{1}{2} \cdot n m \cdot n v^2$$

$$= \frac{1}{2} \times 40^2 \cdot m$$

$$= \frac{1}{2} \times 1600 \cdot m$$

$$= 800 \cdot m$$

$$K.E. = \frac{1}{2} \cdot m \cdot v^2$$

$$\frac{n K.E.}{K.E.} = \frac{800 \cdot m}{\underline{20} \cdot m} \quad \times$$

$$= \frac{800}{\underline{20}}$$

$$= 40 \text{ m}$$

$$\frac{n K.E.}{K.E.} = \frac{\overset{16}{\cancel{800}}}{\cancel{50}}$$

$$= 16$$

$$(13) \quad F = \frac{G m_1 m_2}{r^2}$$

$$r_a = 2 \times r_b$$

Surface object mass = x

$$F_a = \frac{G \cdot M_a \cdot x}{(2 \cdot r)^2}$$

$$= \frac{G \cdot M_a \cdot x}{4 \cdot r^2}$$

$$F_b = \frac{G \cdot M_b \cdot x}{r^2}$$

$$\frac{F_a}{F_b} = \frac{\frac{G \cdot M_a \cdot x}{4 \cdot r^2}}{\frac{G \cdot M_b \cdot x}{r^2}}$$

$$= \frac{G \cdot M_a \cdot x}{4 \cdot r^2} \times \frac{r^2}{G \cdot M_b \cdot x}$$

$$= \frac{1}{4} \times \frac{G \cdot M_a \cdot x}{G \cdot M_b \cdot x} \times \frac{r^2}{r^2}$$

$$= \frac{1}{4}$$

Planet B has stronger gravity at its surface. ✓

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$$P \cdot V = n \cdot R \cdot T$$

$$nV = V$$

$$nP = ?$$

$$nT = 2 \cdot T$$

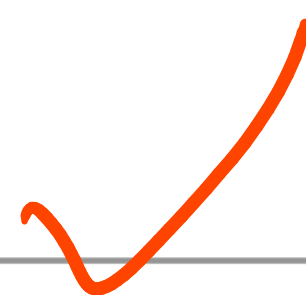
$$nP \cdot nV = n \cdot R \cdot nT$$

$$nP = \frac{n \cdot R \cdot 2 \cdot T}{V}$$

$$= \frac{2 \cdot n \cdot R \cdot T}{V}$$

$$= \frac{2 \cdot PV}{V}$$

$$nP = 2P$$



Variable names are
confusing!!

Use $P, V, T,$
 P_2, V_2, T_2

(15)

$$P \cdot V = n \cdot R \cdot T$$

$$n P = 2 P$$

$$n V = ?$$

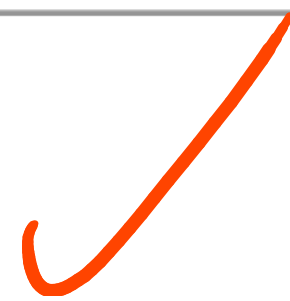
$$n T = T$$

$$n P \cdot n V = n \cdot R \cdot n T$$

$$n V = \frac{n \cdot R \cdot T}{2 P}$$

$$= \frac{P V}{2 P}$$

$$= \frac{V}{2}$$



(16)

I need help

$$(\text{stress}) = \frac{\sigma}{\text{young's modulus}} = \frac{Y \cdot \epsilon}{\text{modulus}} \quad (\text{no constant})$$

stretch 's' $\propto$ amount of pull 'p'
 $\frac{s}{p} = k$

$$s_1 = 1 \text{ mm} ; p_1 = 5 \text{ kg}$$

$$s_2 = ? ; p_2 = 15 \text{ kg}$$

$$\frac{s_1}{p_1} = \frac{s_2}{p_2} \Rightarrow s_2 = \frac{s_1}{p_1} \times p_2$$

$$= \frac{1}{5} \times 15 = 3 \text{ mm}$$

⑦

I need help

Strain (ϵ) is like how much the string stretches

Stress (σ) is like how much force you are applying on the spring.

$$\sigma = Y \cdot \epsilon \quad (Y \text{ is constant})$$

$$\sigma_B = ? ; \epsilon_B = \epsilon_A ; Y_A = 2 \times Y_B$$

$$\sigma_B = Y_B \cdot \epsilon_B = \frac{Y_A}{2} \times \epsilon_A = \frac{\sigma_A}{2}$$

$\therefore \sigma_A$ is higher.
(Therefore) So A needs more stress.

①⑧ mass = m

$$v = v$$

$$r = 50 \text{ m}$$

$$F = \frac{m \cdot v^2}{50}$$

$$m = m$$

$$v = 2 \cdot v$$

$$r = r$$

$$F = \frac{m \cdot (2 \cdot v)^2}{50}$$

$$= \frac{m \cdot 4v^2}{50}$$

$$F = \frac{mv^2}{r}$$

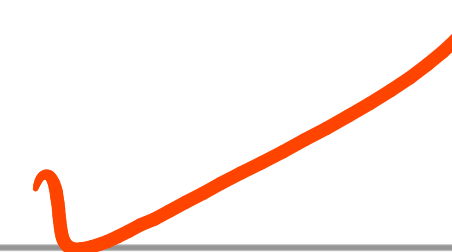
$$\frac{F}{F} = \frac{\left(\frac{m \cdot 4v^2}{50} \right)}{\left(\frac{m \cdot v^2}{50} \right)}$$

$$= \frac{m \cdot 4v^2}{50} \times \frac{50}{m \cdot v^2}$$

$$= 4 \times \frac{50}{50} \times \frac{mv^2}{mv^2}$$

$$\frac{F}{F} = 4$$

$$F = 4 \times F$$



$$(19) \quad F = \frac{mv^2}{r}$$

$$n\check{v} = 2v$$

$$nF = F \approx \frac{mv^2}{r}$$

$$nr = ?$$

$$nm = m$$

$$nF = \frac{nm \cdot nv^2}{nr}$$

$$nF \cdot nr = \frac{nm \cdot nv^2}{nr} \times nr$$

$$= nm \cdot nv^2 \cdot \frac{nr}{nr}$$

$$nF \cdot nr = nm \cdot nv^2$$

$$nr = \frac{nm \cdot nv^2}{nF}$$

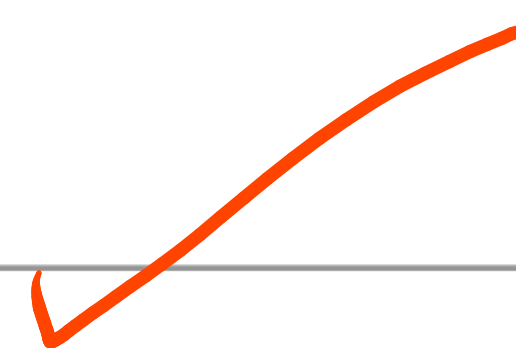
$$nr = \frac{nm \cdot nv^2}{nF}$$

$$= \frac{m \cdot (2v)^2}{\frac{mv^2}{r}}$$

$$= \frac{m \cdot 4v^2}{1} \times \frac{r}{mv^2}$$

$$= \frac{4 \cdot r \cdot m \cdot v^2}{mv^2}$$

$$= 4 \cdot r$$



$$\textcircled{20} F = \frac{mv^2}{r} \quad (\text{Formula})$$

$$F_{\text{car}} = \frac{m_{\text{car}} \cdot v^2}{r}$$

$$F_{\text{truck}} = \frac{m_{\text{truck}} \cdot v^2}{r}$$

$$\frac{F_{\text{truck}}}{F_{\text{car}}} = \frac{\left(\frac{m_{\text{truck}} \cdot v^2}{r} \right)}{\frac{m_{\text{car}} \cdot v^2}{r}}$$

$$= \frac{m_{\text{truck}} \cdot v^2}{r} \times \frac{r}{m_{\text{car}} \cdot v^2}$$

$$= \frac{m_{\text{truck}}}{m_{\text{car}}} \times \frac{v^2}{v^2} \times \frac{r}{r}$$

$$= \frac{m_{\text{truck}}}{m_{\text{car}}} \left(\frac{m_{\text{truck}}}{m_{\text{car}}} \right)$$

$$m_{\text{truck}} > m_{\text{car}}$$

So the truck needs more force to turn.